

P425
ADVANCED MATHEMATICS
Paper 2 AND 1
Jan./Feb. 2026
MERRYLAND HIGH SCHOOL ENTEBBE

Name:..... Student No:.....
Stream:.....

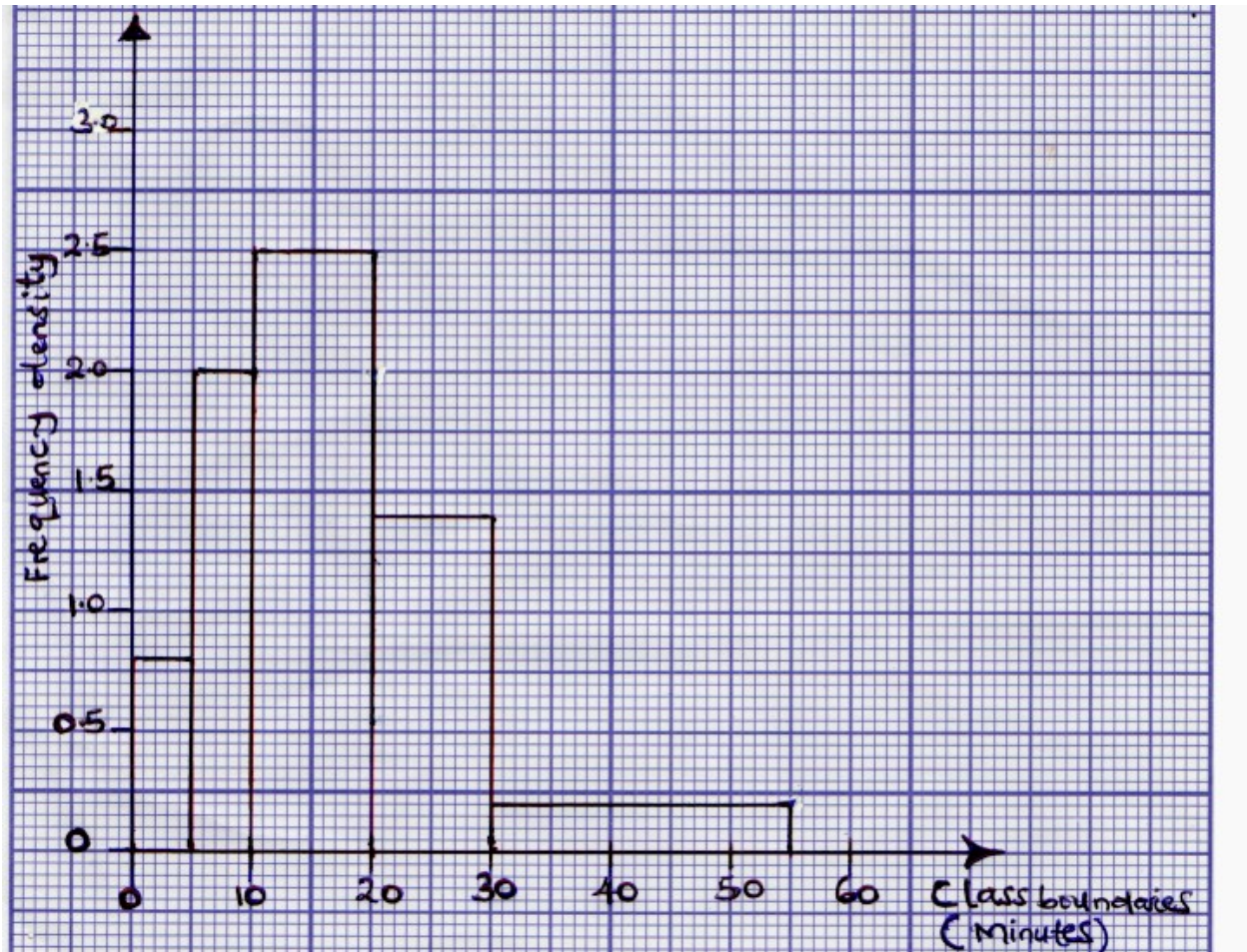
Uganda Advanced Certificate of Education
S.6 APPLIED AND PURE MATHEMATICS
Paper 2 , 1
HOLIDAY PACKAGE

INSTRUCTIONS TO STUDENTS

- Attempt **All items** in this package.
- All necessary working **must** be clearly shown.
- Cancel any work that is not intended for marking.
- If you attempt a question more than once, **only the first attempt** will be marked.
- Number your responses according to the questions attempted.
- Non-programmable calculators and mathematical tables may be used.

ITEM 1

A transport research team studied how long (in minutes) students at Merryland High School spend commuting to school each morning. The results were grouped into travel-time intervals and represented on the stepped diagram below.



Tasks

- From the stepped diagram, Help the team to identify the time interval whose bar is unique above all others.
- Help the team to interpret the information on the diagram into a well-structured statistical way showing all the the important information.
- Assume the school leadership wants to design a fair morning teaching schedule based on actual travel realities.

Using your statistical organization from (b):

- Help the school leaders to determine the overall general duration learners take to reach school.

- (ii) By drawing a progressive graphical form, help the leaders to describe the middle travel duration when learners are arranged from the quickest to the slowest.
- (d) Again help them to determine the Range the duration below which the slowest quarter of learners fall and the duration below which the slowest three-quarters fall.

Item 2

During a quality-sorting exercise, two containers — **Unit Box Alpha** and **Unit Box Beta** — hold small tracking tokens used in a warehouse. The colours of the tokens indicate their processing category:

- Box Alpha contains **5 light-coded tokens** and **2 dark-coded tokens**.
- Box Beta contains **2 light-coded tokens** and **6 dark-coded tokens**.

A technician first retrieves one token from **Box Alpha** without looking and transfers it directly into **Box Beta**. After the transfer, another token is drawn from Box Beta for inspection.

You are required to analyse this two-step process using a **branching probability diagram**.

Tasks

- (a) Using a clearly-labelled branching diagram, Help to carry out investigation to determine:
 - (i) The likelihood that both the transferred token and the inspected token belong to the light-coded category.
 - (ii) The likelihood that the token finally inspected from Box Beta belongs to the light-coded category.
 - (iii) The likelihood that the transferred token was dark-coded, given that the inspected token was light-coded.
- (b) From your branching diagram:
 - (i) Find the chance that exactly one of the two tokens involved is light-coded.
 - (ii) Find the chance that both tokens are light-coded.
 - (iii) Find the chance that none of the tokens is light-coded.
- (c) Let the random variable X represent the number of light-coded tokens involved in the complete two-step procedure.
 - (i) Construct a value-probability table for X .
 - (ii) Use this table to compute the expected numerical value of X ,

Item 3

Isaac enjoys adventure games. In his favorite treasure hunt, there are **five hidden treasures** in a virtual forest. The number of treasures Isaac finds in a single run is represented by a **random outcome** X , whose probabilities are given as:

$$P(X = x) = \begin{cases} \frac{1}{x+2}, & x = 1, 2, 3, 4 \\ k, & x = 5 \\ 0, & \text{otherwise} \end{cases}$$

Tasks

- (a) He needs your help to let him know the value of the unknown constant k and hence the number of treasures Isaac is most likely to collect in a single run .
- (b) He also wants to know the average number of treasures he can collect if he plays the game many times. Also, measure how much the number of treasures **usually deviates** from this average.
- (c) Isaac is feeling confident today. Help him to know the **chance that he collects more than 2 treasures** in one run.
- (d) Each treasure earns Isaac **100 points**, but he starts with a **penalty of 50 points**. Let the total points he scores be $Y = 100X - 50$. Determine the **typical score Isaac can expect** and how much his score **usually varies** from this typical score.

Item 4

A small engineering company is designing a custom spring-loaded toy car. The motion of the car depends on the compression of the spring, which is modeled by the cubic equation:

$$2x^3 - 5x + 2 = 0$$

where x represents the **compression distance (in m)** of the spring. For safety and performance, the engineers need to know the points where the spring is in equilibrium.

Tasks:

(a)

The engineers want to determine all possible compression distances that satisfy the equilibrium condition.

- Using a graphical approach, they want your help identify that the equation

$$2x^3 - 5x + 2 = 0$$

has three solutions in the range $-2 < x < 2$.

(b)

The engineers are particularly interested in the root that lies between 1 m and 2 m, as it represents the spring compression for optimal performance of the toy car.

- Use the Newton-Raphson formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

to identify this root.

Item 5

A civil engineer is analyzing the forces acting on a **rectangular steel frame** ABCD used in a playground structure. The frame has dimensions $AB = 4$ cm and $BC = 3$ cm. Various forces act along the sides and the diagonal of the frame as follows:

- A force of 2 N is applied along the side whose direction follows the path from A to the next vertex when moving around the rectangle in alphabetical order.
- A force of 1 N acts along the side that continues the same alphabetical journey beginning from B .
- A force of 5 N acts along the side reached after completing the previous two steps, linking the next vertex to D .
- A force of 6 N is applied along the side that directly connects A with the vertex that comes before A when tracing the rectangle anticlockwise.
- A force of 7 N acts along the internal line that joins the first vertex A with the one diagonally positioned across the rectangle, C .

The engineer wants your help to determine the combined effect of all these forces on the frame.

Tasks:

(a)

- Using vector resolution along horizontal and vertical directions, help the engineer to identify the single force acting on the frame.
- Hence deduce the inclination of the resultant relative to the horizontal side AB .

(b)

The engineer is particularly interested in the effect of the resultant along the **horizontal plane** for placing support beams:

- Construct an expression which relates the action of this horizontal component using the frame's coordinates.
- Interpret the expression : explain how the action informs where supports should be positioned to maintain stability.

(Item 6)

At Merryland High School, the Mathematics Club is planning to construct a **mini water fountain** in front of the administration block. The fountain's design involves a curved water pipe whose shape is modeled by a cubic polynomial. The height of the water stream (in meters) above the ground is given by:

$$h(x) = x^3 - 13x + p$$

where x represents the distance (in meters) along the pipe from the base, and p is a design constant that depends on the water pressure setting.

During the testing, the water jet was observed to pass exactly through a spray point located 4 m from the base.

Task(2)

a) Using the given information, determine the value of p that ensures the pipe passes through the spray point.

b) Hence, find the other points along the pipe where the water level is at ground level, that is, where $h(x) = 0$.

To ensure safety, the school wants to install a **metal guard rail** around the fountain area. The safety rule requires that the ratio of the **height (in cm)** of the rail to its **distance from the fountain** satisfies the condition:

$$\frac{x^2 - x - 18}{x + 3} \geq \frac{x}{2}$$

where x represents the distance (in meters) from the fountain.

Task(2)

a) Solve the inequality to determine the safe range of distances (x) within which the guard rail can be installed.

b) Suggest the most suitable position for the guard rail that ensures safety and efficient use of space around the fountain.

Item 7

During the renovation of School's compound, the school architect designed a **triangular flower garden** in front of the wood land block. The garden is to be fenced along three

straight paths that meet at different points. The coordinates of two of these points on the school compound plan are:

$$A(0, -2) \text{ and } B(4, 0).$$

The third point, $C(h, k)$, represents the tip of the triangle where a decorative fountain will be placed. Part of the line with equation

$$4x - 3y = 6$$

forms the fence line between the fountain and point A (side AC), while part of the line with equation

$$3x + 4y = 12$$

forms the fence line between the fountain and point B (side CB).

Tasks:

- a) Determine the coordinates (h, k) of the position where the fountain will be constructed.
- b) Find the shortest distance from the fountain to the line segment joining points A and B .
- c) Interpret your result in the context of the school plan.
- d) The school intends to plant flowers symmetrically along the two sides AC and CB . Determine the equations of the **bisectors of the angles** between the two fence lines, which will guide the planting design.

Item 8

The School Science Club is working on two classroom demonstrations that require the use of algebraic models. **Demonstration A (Balance Sensor):**

A small electronic balance used in the physics laboratory is modelled by the quadratic equation

$$ax^2 + bx + c = 0,$$

where the two roots x_1 and x_2 represent calibrated deflection settings at which the device reads zero. During calibration, the technicians observe an interesting property: the **sum of the roots** equals the **sum of the squares of their reciprocals**. **Demonstration B**

(Concentration Indicator):

In the chemistry corner, students are testing two connected sensors for measuring solution concentration. The first sensor reading, denoted by t , is given by

$$t = \sqrt{\frac{x-1}{3x+2}},$$

where x is a concentration parameter to be determined. The second sensor produces a reading equal to $\frac{2}{t}$.

During the calibration experiment, the sum of the two readings was recorded as 3.

Tasks:

If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares of their

reciprocals, show that the coefficients satisfy the relationship:

$$ab^2 + bc^2 = 2a^2c.$$

Using the information from Demonstration B:

1. Let t be the first sensor reading and the second reading be $\frac{2}{t}$. Write down the equation that models the calibration condition that the sum of readings equals 3.
2. Using the substitution $t = \sqrt{\frac{x-1}{3x+2}}$, solve for all real values of x that satisfy this condition.
3. State and justify any domain restrictions for x , and verify your final answers in the original context.

Item 9

The *Robotics Club* is designing a **signal transmission circuit** for a remote-controlled science project. In testing, the club discovered that certain electrical signals can be represented using **complex numbers**, which capture both the *magnitude* (strength of the signal) and the *phase angle* (timing shift). During the experiment, one signal was represented by the complex expression:

$$z = \frac{(1-i)(\sqrt{3}-i)}{1-i\sqrt{3}}.$$

The team needs to analyze this signal to understand its behavior and effects on the circuit.

Task

1. The technician explains that every complex signal z can be written in *polar form* as

$$z = r(\cos \theta + i \sin \theta),$$

where r is the magnitude and θ is the argument (phase angle). Using this information, express z in polar form.

2. To visualize the phase relationship between signals, the Robotics Club plots each signal on an *Argand diagram*. Draw a neat Argand diagram showing the position of z , and clearly label its magnitude and argument.
3. A problem occurs in the circuit: the signal response seems unstable. The club's mentor asks for your help to find the possible **response signals** that satisfy the equation

$$z^2 - 4(1+i)z + 9 + 8i = 0.$$

Use your knowledge of complex numbers to help the Robotics Club determine:

- (a) The two possible signal values of z .

Item 10

Maryam runs a small printing business that produces posters of various widths. The ink consumption per poster, $I(x)$ (in liters), depends on the width x (in meters) according to the formula:

$$I(x) = \frac{2x^2 - 5x + 7}{(4x^2 - 9)(x + 2)}$$

Maryam is planning her ink orders for the next month. However, she has limited storage space and wants to **avoid over-ordering or under-ordering ink**. Directly using the formula for multiple poster sizes is cumbersome.

Tasks

- Express $I(x)$ as a sum of simpler fractions.
- Most of Maryam's posters are less than 1 meter wide approximate $I(x)$ as a **simple polynomial in ascending powers of x** up to the x^2 term.
- Determine the **rate at which ink consumption changes** with respect to width x .

Item 11

James is designing a triangular garden in his backyard. The garden has pathways along each side, and he needs to calculate angles and lengths accurately for landscaping and decorations.

1. The slope of one pathway is such that the relation between its incline angle x (in radians) and safety measures is given by:

$$7 \cot x - \csc x = 5, \quad 0 \leq x \leq \frac{3}{2}\pi$$

James wants to determine the possible angles x to ensure both safety and aesthetic design.

2. For the main triangular flowerbed PQR, he knows:

- Two sides measure 5 m and 7 m.
- The included angle between them is 45° .

He also knows that in any triangle:

$$\tan \frac{Q - R}{2} = \frac{q - r}{q + r} \cdot \cot \frac{P}{2}$$

He wants to use this relation to determine all the unknown angles and the third side in his garden triangle.

Tasks

- Solve the equation

$$7 \cot x - \operatorname{cosec} x = 5$$

for x in the interval $0 \leq x \leq \frac{3}{2}\pi$.

- Discuss which solution(s) are **physically realistic** for the slope of a pathway.
- Using the triangle PQR and the given relation, **show that**

$$\tan \frac{(Q - R)}{2} = \frac{q - r}{q + r} \cdot \cot \frac{P}{2}.$$

- Solve the triangle to find all unknown sides and angles.
- Explain how these values would help James **plan the flowerbed layout** effectively.

Item 12 A school is organizing different activities for its annual math festival. The organizers have several tasks to complete and need your help with calculations. Tasks.

(a): The festival committee has n volunteers. They want to select m volunteers for a special team. One of the committee members suggests two possible methods to count the selections:

1. Selecting $m - 1$ volunteers from the remaining $n - 1$ volunteers and then adding the team leader.
2. Selecting m volunteers from the remaining $n - 1$ volunteers and including the team leader separately.

Show that the total number of ways using these two methods is equal to $\binom{n}{m}$, i.e.,

$$\frac{(n-1)!}{(n-m)!(m-1)!} + \frac{(n-1)!}{(n-m-1)!m!} = \binom{n}{m}.$$

(b) The festival committee wants to create a banner using the word **MATHEMATICIANS**. They want to arrange the letters such that **the two M's are never next to each other**.

Determine the number of possible arrangements satisfying this condition.

- (c) If $\frac{1}{(b+c)}$, $\frac{1}{(a+c)}$ and $\frac{1}{(a+b)}$ are consecutive terms of an arithmetic progression, prove that a^2 , b^2 and c^2 are also three consecutive terms of an arithmetic progression.

Use this holiday to sharpen your problem-solving skills by reviewing key concepts a little each day so you return confident and ready for greater challenges.